

Differential Diagnosis of Memory Disorders

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Session Objectives

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- 1) Understand how to evaluate memory loss and compare dementia to delirium.
- 2) Describe proper history taking in the setting of memory loss.
- 3) Generate a differential diagnosis for memory loss.
- 4) Consider reversible causes of memory loss.
- 5) Apply appropriate treatment options (if they exist).

Mystery Case

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- Alfredo was a 53-year-old healthy Italian man.
- While on the cruise ship's dance floor, he became drenched in sweat.
- He checked the mirror- his pupils were pinpoint bilaterally.
- Then he felt a “shock”- like he put his finger in an electric socket.

The Family that Couldn't sleep DT Max



Take a good history

- Details:
- 1) Time course- slow or rapid (years or months?)
- 2) Trauma?
- 3) Fever or infection?
- 4) Toxic habits? HIV risk factors?
- 5) Vascular risk factors?
- 6) Other illnesses?

Delirium

- Disturbance of consciousness with reduced ability to focus
- Develops over a short period of time (hours-days)
- Fluctuates during the course of the day. There is an additional disturbance in memory, disorganization, language, visuospatial ability or perception.
- Generally reversible

Dementia

- Follows gradual course- months to years.
- With or without confusion, but disturbance in level of consciousness is generally not a feature.
- Dementia is an independent risk factor for delirium.

Diagnostic and Statistical Manual of Mental Disorders- 5th Edition
Image: Official portrait

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Irreversible Causes of Dementia

- Early or late onset Alzheimer's Disease
- Frontotemporal Dementia
- Chronic traumatic encephalopathy
- Lewy Body Dementias
- Vascular Cognitive Impairment
- Cerebral Amyloid Angiopathy
- Creutzfeld-Jakob, Prion disease

Reversible Causes of Dementia

- Hypothyroidism
- Vitamin B1 or B12 Deficiency
- Depression
- Seizures
- Normal Pressure Hydrocephalus
- Subdural Hematoma
- Infection
- Brain Tumor
- Intracranial hypotension
- Medications, alcohol or drug dependency

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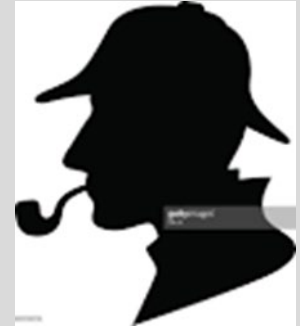
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Mystery Case

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- Alfredo began to note changes in his memory and attention.
- Sleep was affected in that he never felt refreshed.



The family that couldn't sleep, by DT Max

Mental Status Evaluation

- Level of alertness
- Orientation- to self, date, place
- Language- naming high and low frequency words
- Working memory- repeat digit span (7+/-2)
- Complex attention- repeat digit span or months in reverse order

Case 1

- 75-year-old man
- Several year history of memory loss.
- Asks the same question repeatedly
- No longer balancing his checkbook x 1 year.
- Continues to feed and dress himself.
- Neurologic exam is non-focal.
- Labs and imaging are normal.

MCI (mild cognitive impairment)

- In-between state of normal aging and dementia.
- MCI cognitive change > than expected for age.
- Independence in the community and ADLs are preserved.



Alzheimer's Disease

- 47-55 million people worldwide with AD.
- Most common neurodegenerative disorder.
- 1906- lecture: “On an Unusual Illness of the Cerebral Cortex”- described a 51 year old woman.

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Risk factors for Alzheimer's Disease

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- Age is greatest risk factor
- The presence of 1 or more alleles (APOE4).
- Low educational and occupational attainment.
- Participation in cognitive activities (protective)
- Midlife exercise may decrease dementia likelihood
- Vascular risk factors (HTN, CABG, OH, DM, Obesity)
- Head trauma
- Stress

Clinical presentation (cognitive)

- Changes in verbal and nonverbal memory (asking the same questions and misplacing items)
- There may be difficulties with driving a car or managing complex paperwork.
- MoCA (Montreal Cognitive Assessment) is more sensitive and specific than MMSE in detecting the earliest stages of cognitive impairment.

Clinical Presentation- (psychology)

- Anxiety
- Depression
- Social withdrawal
- Irritability
- Hallucinations (advanced)
- Apathy (advanced)

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Genetics of Alzheimer's Disease

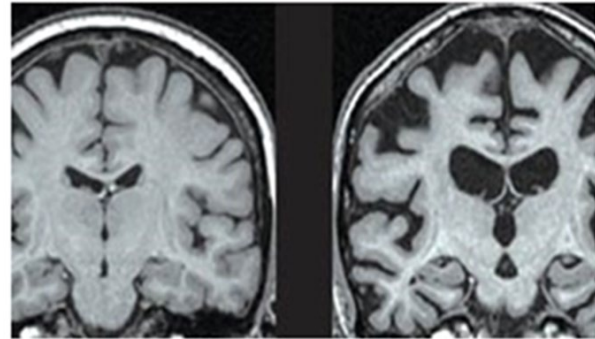
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- **APOE E4** gene variant- most established genetic risk factor for AD
- Early onset Alzheimer's -3 known AD mutations:
- Amyloid precursor protein (**APP**),
- Presenilin 1 (**PSEN 1**)
- Presenilin 2 (**PSEN 2**), (small % of patients).

Imaging

- Neuroimaging is used to exclude structural causes of dementia- including large strokes, brain tumors and infectious causes.
- MRI: progressive increased hippocampal volume loss over time.
- FDG-PET: glucose metabolism is measured with hypometabolism seen in the tempoparietal region in AD



Treatment of AD

- Acetylcholinesterase inhibitors (donepezil and rivastigmine- if no significant bradycardia or GI side effects).
- NMDA antagonist (memantine) have been the mainstay of treatment for dementia.
- Nonpharmacologic therapies: regular sleep wake cycle, exercise, reduction of sensory stimulation.

Anti-amyloid Therapies

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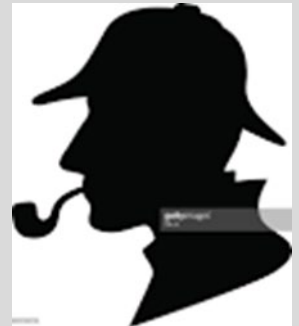
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- Targets amyloid plaques to prevent build-up.
 - Aducanamab (first approved-abandoned 1/2024)
 - Lecanemab-Infusion q 2 weeks
 - Donanemab-infusion q month
 - Adverse effects: ARIA (amyloid related imaging abnormalities)- bleeding and swelling

Mystery Case

- Over the following months Alfredo developed tremors, impotence and severe constipation.
- There were fevers of unclear cause.
- Blood work and imaging were all unrevealing.

The family that Couldn't Sleep DT MAX



Case 2

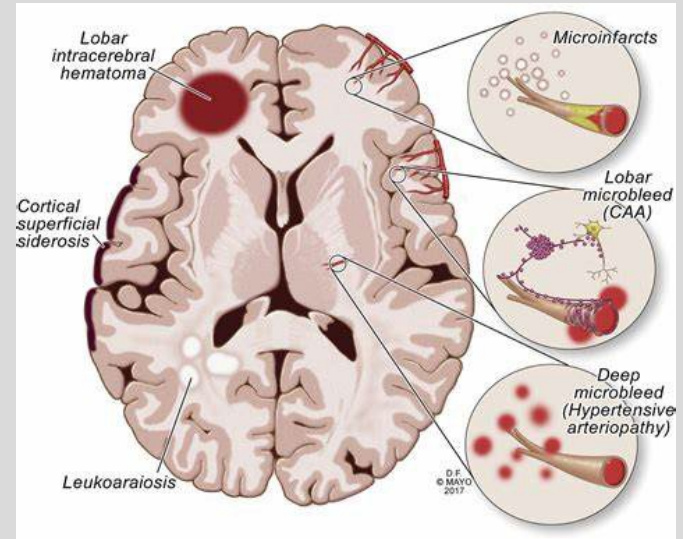
- 70-year-old woman with hypertension referred for evaluation of memory loss.
- Cognitive changes for 3 years prior to presentation (unable to manage accounts at work and lost driving).
- Transient focal neurologic episodes (<30 minutes): motor, sensory, language or visual function- (AKA “amyloid spells”).
 - Paresthesias of the mouth, spreads to other body parts.
- MRI-numerous microhemorrhages

Cerebral Amyloid Angiopathy

- Buildup of amyloid –beta plaques and tau
- APOE4 genotype is major risk factor for CAA.
- CAA is associated with large and small lobar hemorrhages.
- Slowly progressive decline with amyloid spells.

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Vascular Cognitive Impairment and Dementia

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- **Four Phenotypic Categories**
 - Subcortical ischemic vascular dementia- slowed processing speed.
 - Poststroke dementia- history of stroke with language or memory difficulties.
 - Multi-infarct dementia-Multiple large cortical infarcts with apraxia, aphasia, visual field cut or neglect
 - Mixed dementia- vascular cognitive impairment and AD: primarily amnestic syndrome.

Vascular Cognitive Impairment

- AHA/ASA guidelines for optimal brain health
- 1) Manage BP <120/80
- 2) Total cholesterol <200
- 3) Avoid tobacco
- 4) Exercise > 150 min/wk moderate intensity or >75 min wk vigorous
- BMI <25 kg/m²
- Healthy Diet

Case 3

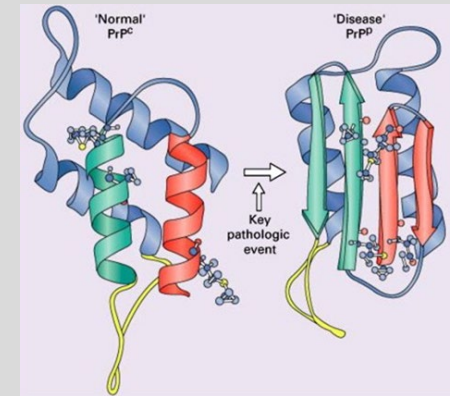
- A 61-year-old healthy man presents with apathy and speaking less.
- 2-3 weeks later he became confused- forgetting why he walked into a room.
- One week later cannot read words on a computer screen or read a clock.
- By end of first month- he could not calculate a tip, stopped driving and working. Occasional myoclonic jerks.

Case 3 continued

- By second month- he spoke in short sentences, used utensils incorrectly and required help to eat and dress.
- By the third month he could no longer be left alone and developed right arm dystonia. Visual deficits worsened- mistook a large bird for a person.
- He died of aspiration pneumonia four months after onset.

Prion Disease

- **Conversion** of the **normal prion protein (PrP^c** prion related protein- c is the cellular form of the protein) It has a helical structure.
- An abnormal form of the protein- called the prion (PrP^{sc} is the final product). Sc is for scrapie, the prion disease of sheep and goats).
- It has a B-pleated sheet structure.



Revised Criteria for CJD

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Core clinical features:

- Rapidly progressive cognitive impairment with 2 or more of the following:
 - Myoclonus
 - Visual or cerebellar disturbance
 - Pyramidal or extrapyramidal signs
 - Akinetic Mutism
 - PLUS imaging and biomarkers

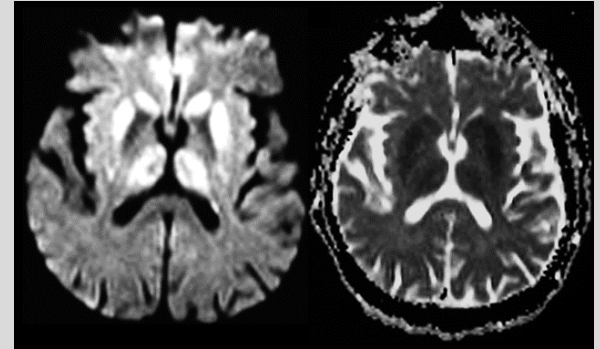


Sporadic Creutzfeldt- Jakob

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- Spongiform Encephalopathy-fatal within one year.
- Sporadic form (85-95%)
- Age of onset: Seventh decade
- MRI- characteristic pattern
- EEG: generalized sharp or sharp-wave complexes
- “Positive” real-time quaking-induced conversion (RT-QuIC) assay for misfolded prion protein and elevated tau (>1150) in the CSF.



Case 4

- 74-year-old man is brought by his spouse
- “You are pretending to be your sister!”
- Significant cognitive “fluctuations”.
- Some days are lucid, other days not. On exam the patient appears parkinsonian
- He has visual hallucinations and is confused.

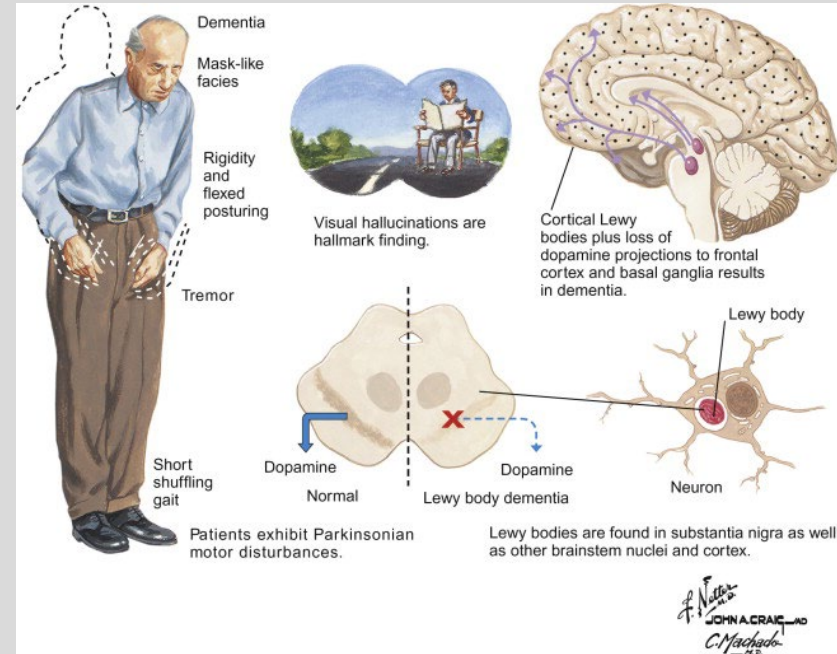


Lewy Body Disease

- Fluctuating cognition or level of consciousness
- Visual hallucinations
- Parkinsonian motor signs
- REM sleep behavior disorder
- Syncope/postural instability
- Apathy, anxiety and depression

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Lewy Body Disease

- Pathology: Intracytoplasmic neuronal inclusions.
- Cortically localized Lewy bodies are strongly related to cognitive impairment and dementia.



Treatment for LBD

- Treatment is symptomatic only.
- Query family regarding safety concerns and driving.
- Sensitivity to meds is common.
- Identify medications that could contribute to clinical worsening- i.e. benzos and anti-cholinergics.
- Cholinesterase inhibitors treat cognitive symptoms and may help with behavioral symptoms.
- Typical antipsychotics are contraindicated.

REM sleep behavior disorder

- Seen in LBD- may occur years before.
- In a healthy physiologic state- vivid mentation is accompanied by skeletal atonia.
- The loss of atonia during REM sleep can result in injurious dream enactment to the patient and/or bed partner.

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Image: Rembehaviorbiopsych.weebly.com. Lifelong Learning in Neurology. Continuum. Sleep Neurology. August 2020, Vol.26, No.4

Mystery Case

- Two months later Alfredo was still having more difficulty sleeping.
- “I haven’t slept in 5 nights”, he exclaimed!
- His wife reported violent jerks at night during sleep.

The family that couldn’t sleep- DT Max



Case 5

- 60-year-old former football player presents with forgetfulness, anger, mood swings and headaches.
- Multiple concussions while playing football.

Case files of A. Leder, DO
Image: sciencemag.org

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Chronic Traumatic Encephalopathy

- Long term sequelae of repetitive mild TBI.
- May occur with contact sports or military blast injuries.
- Secondary to a progressive tauopathy.
- Dementia pugilistica is a subtype of CTE- severe end stage dementia from boxing.

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Image of Dr. Bennet Omalu- health.ucdavis.edu

Continuum Lifelong learning in Neurology. Dementia. February, 2019. Volume 25, No.1

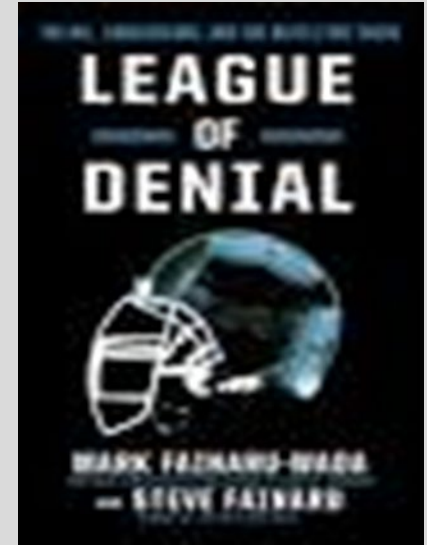
CTE Presentation

- Behavioral
- Cognitive
- Mood Symptoms
- Motor symptomatology



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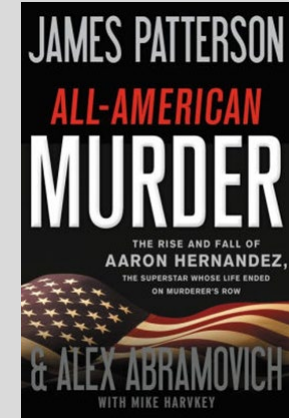
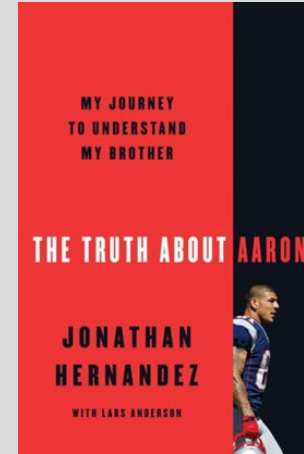
CTE- Mood/Behavioral

- Depression
- Hopelessness
- Explosivity
- Physically violent
- Aggression
- Impulsivity
- Suicidality



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CTE-Cognitive

- Impaired attention/concentration
- Memory loss
- Executive functioning difficulties
- Dementia as the condition progresses

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CTE-Motor

- Dysarthria, dysphagia
- Bradykinesia, tremor
- Rigidity, gait disturbance
- Cerebellar damage
- Falls
- May be coexistent with ALS. “I have atrophy throughout my whole body. And I have this atrophy that I will never recover from in my hands, my arms and now my left leg”- Wesley Walker

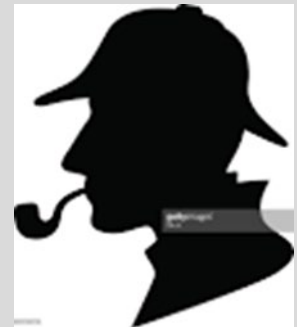
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Mystery Case

- The nights passed for Alfredo- eyes closed, but he was in some type of a trance.
- There were associated hallucinations.
- Hospital personnel believed that he had an alcohol problem, but he denied drinking.



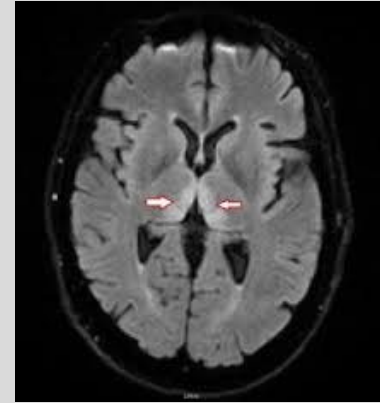
Case 6

- 65-year-old man who smells of alcohol is brought to the ED during your overnight shift.
- He is a poor historian and is confused.
- His eyes were found to be bouncing around.
- He is unable to stand without falling.

Wernicke's Encephalopathy

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- Severe thiamine deficiency (vitamin B1).
- Alcoholics or malnourished patients.
- Classic triad: global confusion, ophthalmoplegia (nystagmus) and ataxia.
- The classic triad not always present. Always have a high level of suspicion.



Wernicke's Encephalopathy

- Korsakoff's psychosis- Memory loss and confabulation (occurs as Wernicke's recedes).
- Lesions in dorsomedial nuclei of the thalamus can cause amnesia.
- Treatment- IV thiamine in the ED. Give this before glucose.

Mystery Case

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- After several months Alfredo began to develop memory loss.
- He was also acting out activities during sleep in a mindless stupor.
- Hallucinations continued. He didn't know if he was alive or dead.

The Family that couldn't sleep- DT MAX



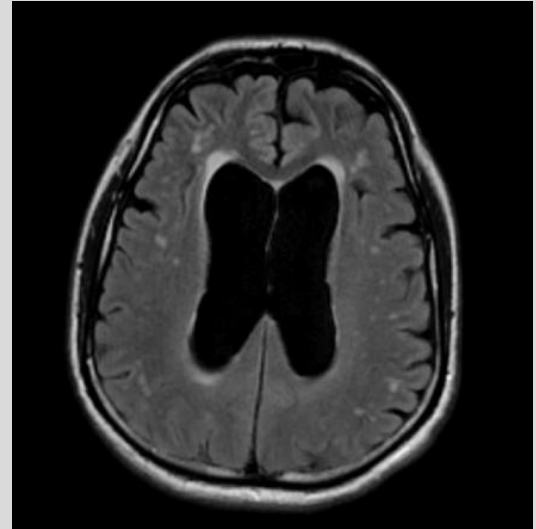
Case 7

- 67-year-old woman with urinary incontinence, gait difficulty and memory loss (classic triad: wet, wobbly, wacky).
- Magnetic gait and appears parkinsonian.
- CT was performed and revealed large ventricles.

Case files of A. Leder, DO
Image: Radiopaedia

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Normal Pressure Hydrocephalus

- Potentially reversible form of memory loss and parkinsonism.
- Gait disturbance and incontinence due to pressure on and distortion of descending white matter pathways.
- Cognitive symptoms due to pressure on anterior cortical structures.

NPH- Treatment

- Videotape gait before the LP
- 30 ml of CSF is drained.
- 30 minutes after LP- observe patient walking again.
- Measure gait, cognition, balance and continence based on specialized scales.
- Determine if VP shunt is the appropriate treatment.

Case 8

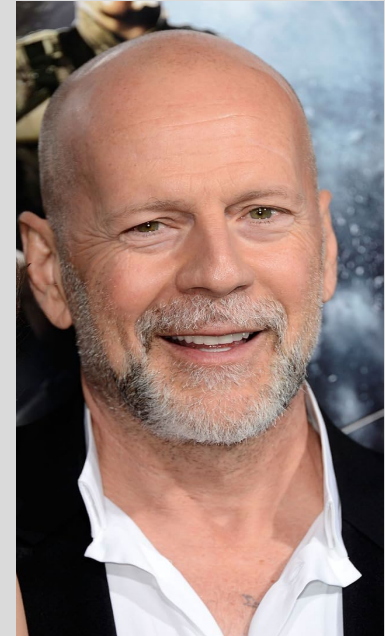
- 51-year-old woman –evaluation for 4 years of behavioral symptoms.
- Was a business executive. Interjected political views, belittled clothing or body habitus of staff members.
- Preoccupied with sugary drinks.
- Starting collecting antique statuettes (\$15,000).
- Refused to drive kids to their games “teams stink anyway”.
- Gifted partly used cigar (found on street) for Father’s Day.

Frontotemporal Dementia

- Specific syndromes under the FTD umbrella include behavior behavioral variant (bvFTD) and primary progressive aphasia (PPA).
- Most cases are sporadic. Familial phenotypes are associated with the *MAPT* gene, *GRN* gene and the *C9orf72* gene.

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Frontotemporal Dementia

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- Clinical Features:
 - Behavioral disinhibition
 - Apathy or inertia
 - Loss of sympathy or empathy
 - Perseverative, stereotyped or compulsive ritualistic behavior.
 - Hyperorality and dietary changes



Frontotemporal Dementia

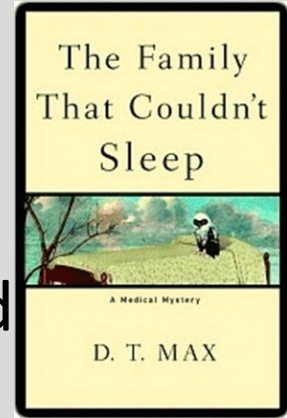
- Diagnosis: Functional neuroimaging (Fludeoxyglucose PET).
- No FDA approved medication at this time.
- Symptoms treated with SSRIs, atypical antipsychotics and trazodone for behavioral dyscontrol and agitation.
- Cholinesterase inhibitors do not improve cognition.

Mystery Case- Revealed!

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- Fatal familial insomnia
- Rare genetic prion disease
- Onset in late 40s
- Severe progressive insomnia over months
- Dysautonomia (tachycardia, hyperhidrosis, and hyperpyrexia).
- Motor and cognitive symptoms occur late.
- 18 months survival



The family that couldn't sleep- DT MAX

Summary Slide

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- Summary Point 1 : Recognize reversible causes of memory loss and don't miss them.
- Summary Point 2 : Understand the underlying cause of memory loss based on the history of the patient.
- Core References:
 - Reference 1: Continuum Lifelong learning in Neurology. Dementia, June 2022, Vol.28, No.3
 - Reference 2: The Family that Couldn't Sleep by DT Max

Lecture and Lab Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>